

FYS5419/9419,
lecture March 26

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$$|x\rangle = \vec{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix}$$

x_j s are complex numbers

$$y_k = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j e^{2\pi i (jk)/n}$$

$$|y\rangle = \vec{y} =$$

$$\frac{1}{\sqrt{n}}$$

$$\begin{bmatrix} \sum_{j=0}^{n-1} x_j' e^{2\pi i j' \cdot 0/n} \\ \sum_{j=0}^{n-1} x_j' e^{2\pi i j' \cdot 1/n} \\ \vdots \\ \sum_{j=0}^{n-1} x_j' e^{2\pi i j' (n-1)/n} \end{bmatrix}$$

$y_0 \quad k=0$
 $y_1 \quad k=1$
 $y_{n-1} \quad k=n-1$

Example

$$\vec{x}^T = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$\nearrow x_0 \quad \nwarrow x_1$

$$k=0 : y_0 = \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}}$$

$$k=1 : y_1 = \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}} \underline{e^{i\pi}}$$

$\quad \quad \quad -1$

$$|y\rangle = U|x\rangle \quad U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$
$$U^\dagger U = U U^\dagger = \underline{1}$$

Example (polynomial expansion)

$$\begin{aligned} f(x_i) &\stackrel{!}{=} P_{n-1}(x_i) = P(x_i) \\ &= \sum_{j=0}^{n-1} a_j x_i^j = a_0 x_i^0 + \\ &\quad a_1 x_i + a_2 x_i^2 \\ &\quad + \dots + a_{n-1} x_i^{n-1} \end{aligned}$$

$$\begin{aligned} |p\rangle &= \begin{bmatrix} p(x_0) \\ p(x_1) \\ \vdots \\ p(x_{n-1}) \end{bmatrix} = V|a\rangle \\ &\quad V \in \mathbb{R}^{n \times n} \end{aligned}$$

$$V = \begin{bmatrix} 1 & x_0^1 & x_0^2 & \dots & x_0^{n-1} \\ 1 & & & & x_1^{n-1} \\ \vdots & & & & \vdots \\ 1 & x_{n-1}^1 & & & x_{n-1}^{n-1} \end{bmatrix}$$

Van der Monde matrix

This is similar to a Fourier transform

$$\omega = e^{2\pi i/n}$$

$$x_0 = e^{2\pi i \cdot 0/n} = \underline{1}$$

$$x_0^2 = e^{2 \cdot 2\pi i \cdot 0 / n} = \underline{1}$$

$$x_1^1 = e^{2\pi i / n} = \omega$$

$$x_1^2 = e^{2\pi i \cdot 2 / n} = \omega^2$$

⋮

$$U = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \dots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \dots & \dots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ \underline{1} & \omega^{n-1} & \dots & \dots & \dots & \omega^{(n-1)^2} \end{bmatrix}$$

Exercise: find u^{-1}

$$(u u^{-1})_{jk} = \frac{1}{n} \sum_{l=0}^{n-1} w^{jl - lk}$$

$$u u^T = u^T u = \underline{1} = \frac{1}{n} \sum_{l=0}^{n-1} w^{(j-k)l}$$

$$w = e^{2\pi i l / n}$$

is equal to 1 only
if $j = k$

QFT

n -qubits; 2^n number of basis states

$$N = 2^n$$

$n = 1 \Rightarrow N = 2^1$; $|0\rangle$ and $|1\rangle$

$n = 2 \Rightarrow N = 4$; $|00\rangle, |01\rangle, |10\rangle$
and $|11\rangle$

$|y\rangle = U|x\rangle$; U preserves orthogonality

$$\langle y_i | y_j \rangle = \langle x_i | \underbrace{u^\dagger u}_{= \delta_{ij}} | x_j \rangle = \delta_{ij}$$

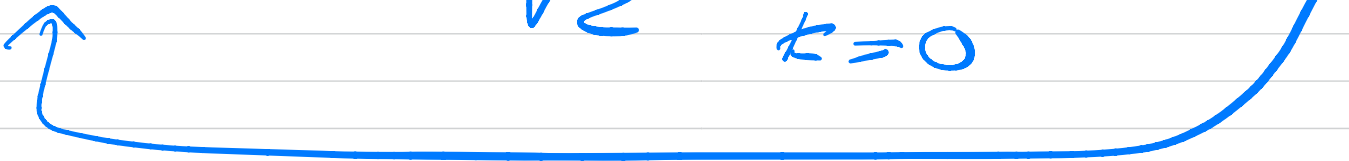
($\langle x_i | x_j \rangle = \delta_{ij}$) $\underline{1}$

QFT is a linear operator that transforms an ONB basis into a new ONB.

Example 1-qubit

$$|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{i2\pi jk/N} |k\rangle$$

$|0\rangle$ and $|1\rangle$

$$|0^z\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^1 e^{2\pi i 0 \cdot k/2} |k\rangle$$


$$= \frac{1}{\sqrt{2}} [|0\rangle + |1\rangle]$$

$$|1^z\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^1 e^{2\pi i 1 \cdot k/2} |k\rangle$$

$$= \frac{1}{\sqrt{2}} [|0\rangle + e^{\pi i} |1\rangle]$$

$$|g\rangle = U|x\rangle$$

$$U = H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & e^{i\pi/4} \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Example: $n=2 \Rightarrow N=2^2=4$

$$|00\rangle \quad |01\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$|10\rangle$ and $|11\rangle$

compact encoding

$$|00\rangle \rightarrow |0\rangle$$

$$|10\rangle \rightarrow |2\rangle$$

$$|01\rangle \rightarrow |1\rangle$$

$$|11\rangle \rightarrow |3\rangle$$

$$|00\rangle \rightarrow |0\rangle$$

$$\hookrightarrow 2^1 |0\rangle + 2^0 |0\rangle = 0$$

$$|01\rangle \rightarrow |1\rangle$$

$$\hookrightarrow 2^1 |0\rangle + 2^0 |1\rangle = 1$$

$$|10\rangle \Rightarrow |2\rangle \Rightarrow 2^1(1) + 2^0(0) = 2$$

$$|11\rangle \Rightarrow |3\rangle \Rightarrow 2^1(1) + 2^0(1) = 3$$

Example $n=3$; $N=2^3=8$

$$|000\rangle, |001\rangle, |010\rangle, \dots, |111\rangle$$

$$|9\rangle = |4\rangle = |100\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2^2(1) + 2^1(0) + 2^0(0) = 4$$

in general we start with

$$|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i j k / N} |k\rangle$$

general basis

$$|k\rangle = \{k_0, k_1, \dots, k_{n-1}\}$$

$$J = 2^{n-1} J_0 + 2^{n-2} J_1 + \dots + 2^1 J_{n-2} + 2^0 J_{n-1}$$

$$|j\rangle = \frac{1}{\sqrt{N}} \sum_{k_0=0}^1 \sum_{k_1=0}^1 \sum_{k_2=0}^1 \dots \sum_{k_{n-1}=0}^1$$

$$\times \left\{ e^{2\pi i j' / N} \sum_{i=0}^{n-1} y_i' z^{n-i} \right\}$$

$$= \frac{1}{\sqrt{N}} \left[e^{i 2\pi j / 2^0} \right] \otimes \left[e^{i 2\pi j' / 2} \right] \otimes \dots \otimes$$

$$\begin{bmatrix} 1 \\ e^{i2\pi j/2^{n-1}} \end{bmatrix}$$

$$|0\rangle + e^{i2\pi j/2^k} |1\rangle =$$

$$\begin{bmatrix} 1 \\ e^{i2\pi j/2^k} \end{bmatrix}$$

Example $n=2$

$$|00\rangle = ?$$

$$\frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 1 \end{bmatrix} =$$

$$|00\rangle \rightarrow \frac{1}{2} [|00\rangle + |01\rangle + |10\rangle + |11\rangle]$$

Example $n=3$

$$|001\rangle \rightarrow \frac{1}{\sqrt{8}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ e^{i2\pi/2} \\ e^{-i2\pi/2} \end{bmatrix}$$

$$= \frac{1}{\sqrt{8}} \begin{bmatrix} 1 \\ i \\ -i \\ -1 \\ i \\ -i \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ e^{i2\pi/4} \\ e^{-i2\pi/4} \end{bmatrix}$$

$$= \frac{1}{\sqrt{8}} \left[|1000\rangle + i|1001\rangle - |1010\rangle \right. \\ \left. - i|1011\rangle + |1101\rangle + i|1100\rangle \right. \\ \left. - |1110\rangle - i|1111\rangle \right]$$

For $n=2$

$$U = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & i & -i \\ 1 & -i & -1 & i \end{bmatrix}$$

in general (QFT $N=2^n$)

$$U = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \dots & \dots & \omega^{(N-1)^2} \end{bmatrix}$$

$$\omega = e^{2\pi i/N}$$

$$|y\rangle = |y_0 y_1 \dots y_{n-1}\rangle \quad (10c17)$$

Transform in terms

$$\frac{1}{\sqrt{N}} \left(\begin{array}{c} |0\rangle + e^{2\pi i y / 2^0} |1\rangle \\ |0\rangle + e^{2\pi i y / 2^1} |1\rangle \\ \vdots \end{array} \right) \otimes$$

$$\otimes \left(\begin{array}{c} |0\rangle + e^{2\pi i y / 2^{n-1}} |1\rangle \end{array} \right)$$

$$y = y_0 2^{n-1} + y_1 2^{n-2} + \dots + y_{n-1} 2^0$$

with binary floats-

$$\textcircled{1}. y_k y_{k+1} \dots y_{n-1} =$$

$$y_k 2^{-1} + y_{k+1} 2^{-2} + \dots + y_{n-1} 2^{-(n-k)}$$

$$n > k$$

Example $n=2$

$$\textcircled{0}. y_1 = y_1/2$$

$$\textcircled{0}. y_0 y_1 = y_0/2 + y_1/4$$

$$|y\rangle = |y_0 y_1\rangle \Rightarrow$$

$$= \frac{1}{2} [|0\rangle + e^{i2\pi 0.5} |1\rangle]$$

$$\otimes [|0\rangle + e^{2\pi i 0.5} |1\rangle]$$

$$|00\rangle \rightarrow \frac{1}{2} [|0\rangle + |1\rangle] \otimes$$

$$[|0\rangle + |1\rangle]$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$|y\rangle = |y_0 y_1 \dots y_{n-1}\rangle \xrightarrow{QFT}$$

$$\frac{1}{\sqrt{2^n}} \left(|0\rangle + e^{i2\pi 0 \cdot y_{n-1}} |1\rangle \right)$$

$$\otimes \left(|0\rangle + e^{i2\pi 0 \cdot y_{n-2} y_{n-1}} |1\rangle \right)$$

⋮

$$\otimes \left(|0\rangle + e^{2\pi i 0 \cdot y_0 y_1 y_2 \dots y_{n-1}} |1\rangle \right)$$